

CDAC-Mumbai
Practical Machine Learning

Date: 07-05-2026

Total Marks:25

Time duration: 1.30 hours.

Batch: Feb26

Ques 1 : Develop a Heart Disease Prediction System using machine learning to predict the presence of heart disease based on patient health data. The system should analyze medical parameters such as age, blood pressure, cholesterol level, maximum heart rate, ECG-related features, and other clinical indicators to support early risk assessment. **[15 marks]**

1. Data Collection and Preprocessing (3 Marks)

- Collect heart disease dataset containing patient medical records
- Handle missing and inconsistent data
- Perform feature encoding if required
- Apply feature scaling using StandardScaler
- Split dataset into training and testing sets

2. Exploratory Data Analysis (2 Marks)

- Analyze patient age, cholesterol, blood pressure, and heart rate patterns
- Create graphs and medical data visualizations
- Identify major factors influencing heart disease prediction

3. Model Development (5 Marks)

Implement and compare different machine learning classification models:

- Logistic Regression
- Random Forest
- Decision Tree
- Support Vector Machine
- K-Nearest Neighbors

4. Performance Evaluation (2 Marks)

- Evaluate model using confusion matrix and classification report
- Compare accuracy, precision, recall, and F1-score
- Identify error patterns in prediction results

5. Result Analysis and Conclusion (3 Marks)

- Interpret prediction outcomes
- Explain the importance of preprocessing and scaling
- Discuss future improvements using advanced models and hyperparameter tuning

Ques 2 : Develop an Employee Attrition Analysis and Cluster Prediction System using machine learning techniques to identify factors affecting employee attrition and group employees into meaningful clusters using supervised and unsupervised learning algorithms. The system should analyze employee demographic, professional, performance, salary, and satisfaction-related data to support HR decision-making and workforce retention strategies. **[10 marks]**

1. Data Collection and Preprocessing (2 Marks)

- Collect employee attrition and performance dataset
- Handle missing values and inconsistent records
- Encode categorical variables
- Perform feature scaling and normalization
- Prepare data for supervised, regression, and unsupervised learning models

2. Exploratory Data Analysis (EDA) and Business Insights (2 Marks)

- Analyze employee demographics, job roles, salary, satisfaction, and work-life balance
- Identify attrition trends using statistical and visual analysis
- Identify the employee turnover
- Create charts, heatmaps, and correlation visualizations

3. Feature Engineering and Attrition Relation Analysis (2 Marks)

- Identify important features influencing employee attrition
- Study relationships between satisfaction, overtime, salary, and attrition
- Perform correlation and feature importance analysis
- Apply Linear Regression to analyze relationships between salary, experience, performance rating, and employee satisfaction
- Generate actionable HR insights for retention strategies

4. Employee Cluster Prediction using Unsupervised Learning (2 Marks)

Implement clustering algorithms to group employees based on similar characteristics:

- K-Means Clustering
- DBSCAN Clustering

Cluster Analysis

- Identify employee segments such as:
 - High-risk attrition groups
 - High performers
 - Dissatisfied employees
 - Long-term loyal employees
- Visualize clusters using dimensionality reduction techniques

6. Model Evaluation, Conclusion, and Recommendations (2 Marks)

- Evaluate regression model using Mean Squared Error (MSE) and R^2 Score
- Analyze clustering quality using silhouette score
- Interpret cluster behavior and attrition patterns
- Provide HR recommendations to reduce employee turnover

----- All the Best !-----